Soil & Nutrients: Reading

Horticulture 1 · Unit 5 · Level 3 (closer to the class work) · goes with Plant Nutrition Definitions, Soil Analysis Parts 1-2, deficiencies

Name: _____

Soil, Media and Plant Nutrients

Plants need light, water, air, the right temperature, and a growing medium that supplies nutrients. In a greenhouse, plants grow in potting mix, a blend of peat or bark with perlite or vermiculite. Potting mix drains well, holds air for roots and is free of weed seeds. Dry mix should be wetted first, because its dust can carry germs.

Garden soil is made of mineral particles of different sizes. Sand particles are the largest and let water drain quickly. Silt particles are medium-sized and feel smooth. Clay particles are the smallest; they are sticky and hold water. Loam is a balanced mix of the three and is ideal for most plants. A soil texture triangle uses the percentages of sand, silt and clay to name a soil type.

In a soil jar test, soil and water are shaken in a clear jar and left to settle. Sand settles first at the bottom, then silt, with clay on top. Measuring each layer and dividing by the total depth, then multiplying by 100, gives the percent of each particle.

The three main nutrients are listed on fertilizer bags as N-P-K. Nitrogen (N) promotes green, leafy growth; a deficiency causes yellowing of older leaves, and an excess causes soft, leafy growth with few flowers. Phosphorus (P) supports roots, flowers and seeds; a deficiency can cause purple leaves. Potassium (K) builds strong stems and disease resistance; a deficiency causes brown, scorched leaf edges.

Check your reading

1. Why do greenhouses use potting mix instead of garden soil?

2. In the jar test, why does sand settle first?

3. Name one deficiency symptom for N, P and K.